Experiment 8 a)

Nmap Live host & Discovery using ARP, ICMP, TCP, UDP

Aim:

To discover live hosts using Nmap scans such as ARP, ICMP, TCP & UDP on the TryHackme platform.

Algorithm:-

1) Open TryHackme & access the Nmap Live Host Discovery room

2) Study the concept of live host discovery.

3) Perform ARP scanning to identify live hosts.

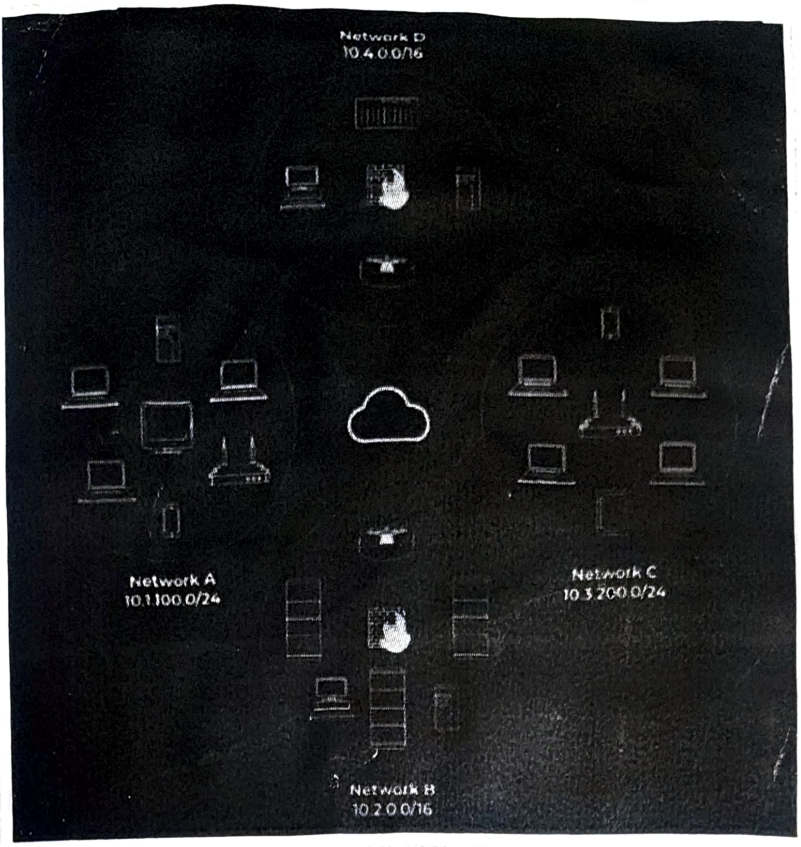
4) Perform ICMP scanning to check host responses

5) Perform TCP SYN & TCP ACK ping scans

6) Perform UDP ping scanning to check host reachability

7) Compare the results of all scanning methods

8) Record the live hosts identified by Nmap.



Send Packet

From:

computer1

To:

computer1

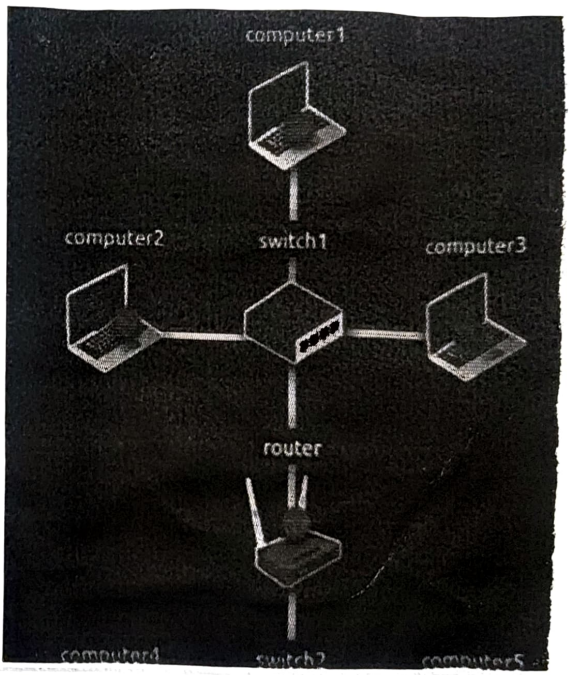
Packet Type:

arp_request

Data:

computer6

Send Packet



Send Packet

From:

computer4

To:

computer4

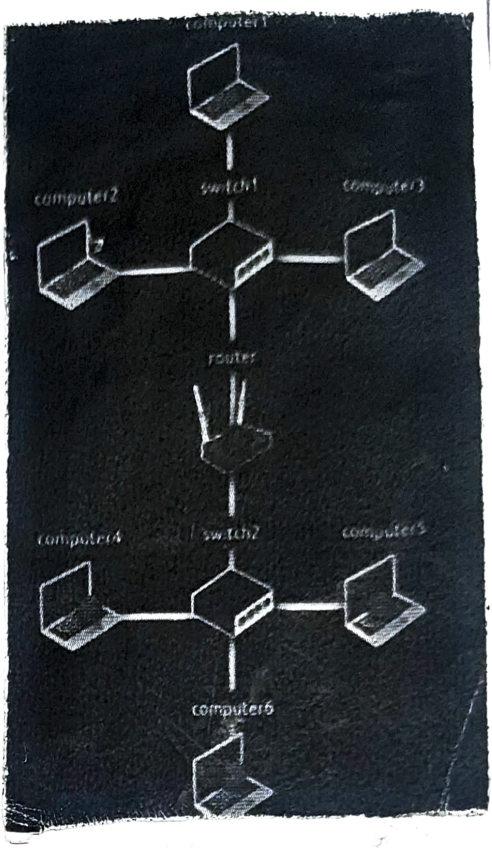
Packet Type:

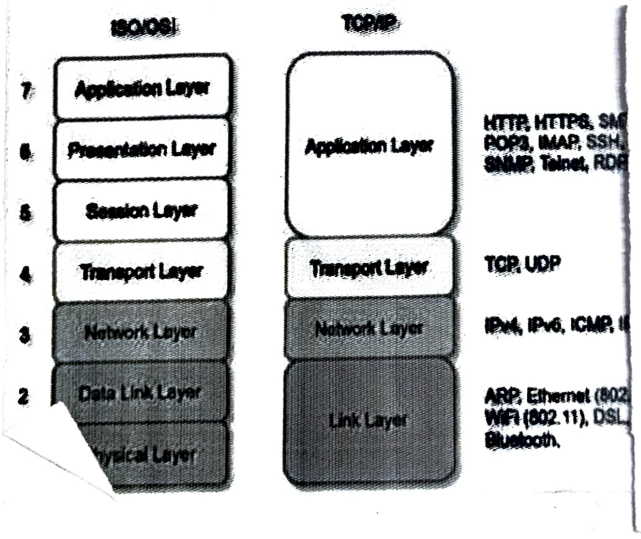
arp_request

Data:

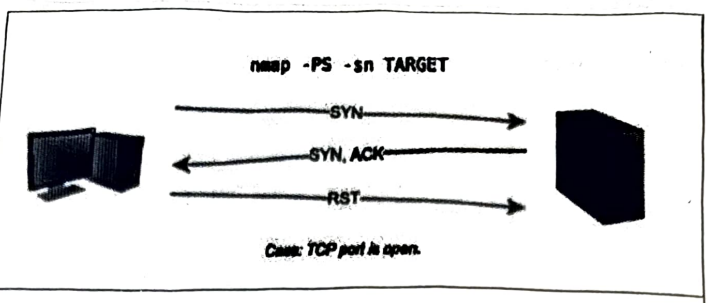
computer6

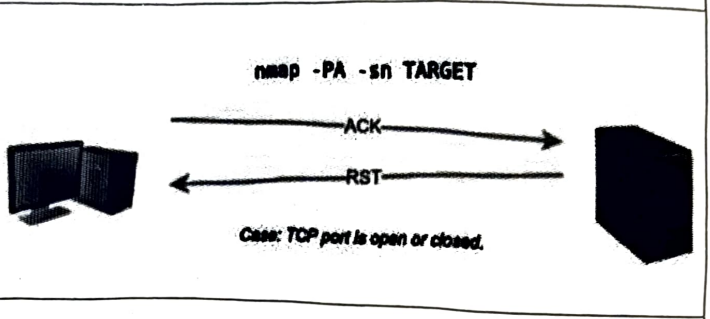
Send Packet

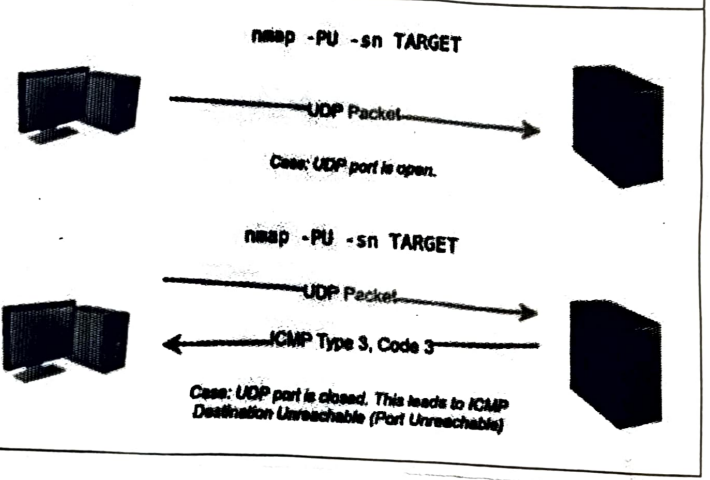


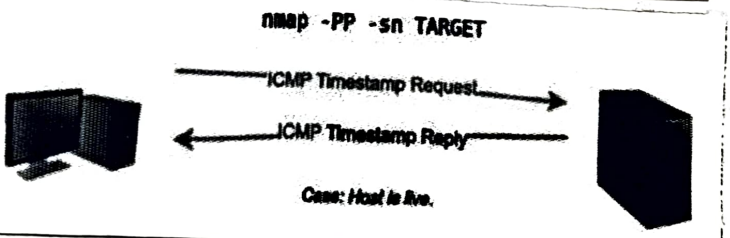


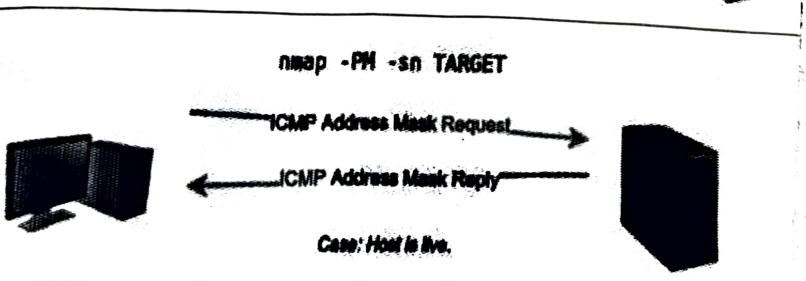


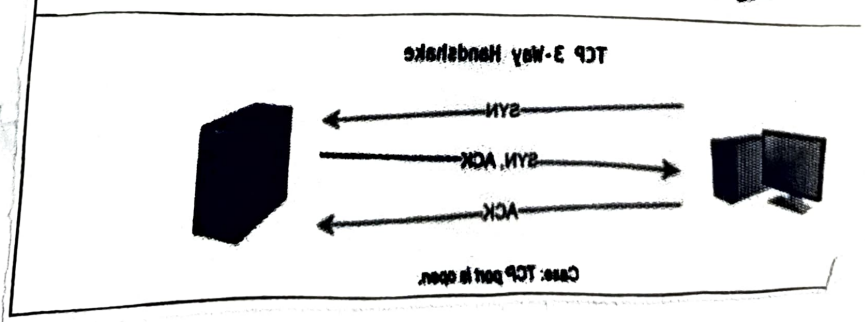












hosts were successfully identified using ARP, ICMP,

Experiment 8b)

Configuration of wireless LAN using cisco Packet Tracker

Aim:

To configure a WLAN using cisco packet tracker by connecting three PCs to a Linksys wireless Router, assigning static IP address, configuring the SSID & WEP.

Algorithm:-

1) Open Cisco Packet Tracker & place a wireless router and three PCs.

2) Configure the router with SSID MotherNetwork and WEP security

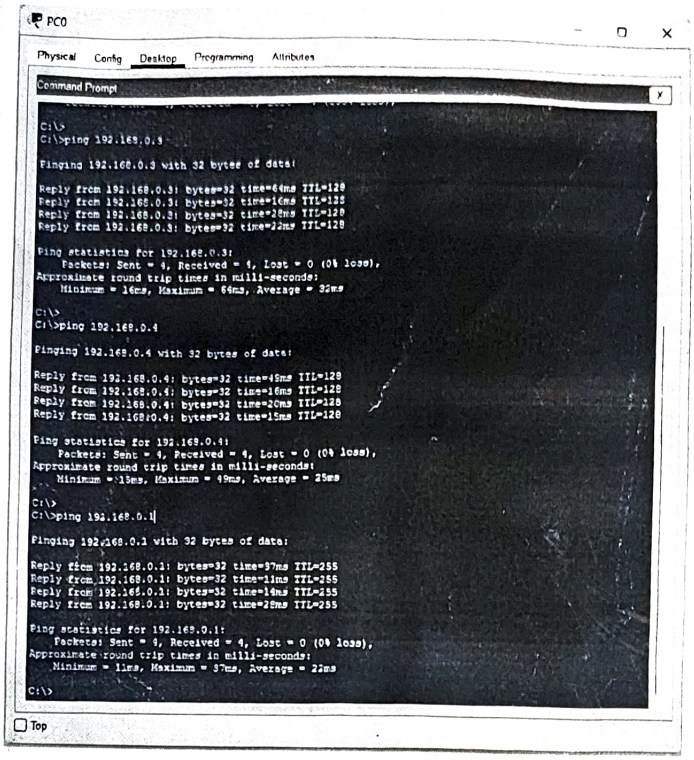
3) Set the static IP addresses of the PCs are 192.168.0.2, 192.168.0.3, 192.168.0.4

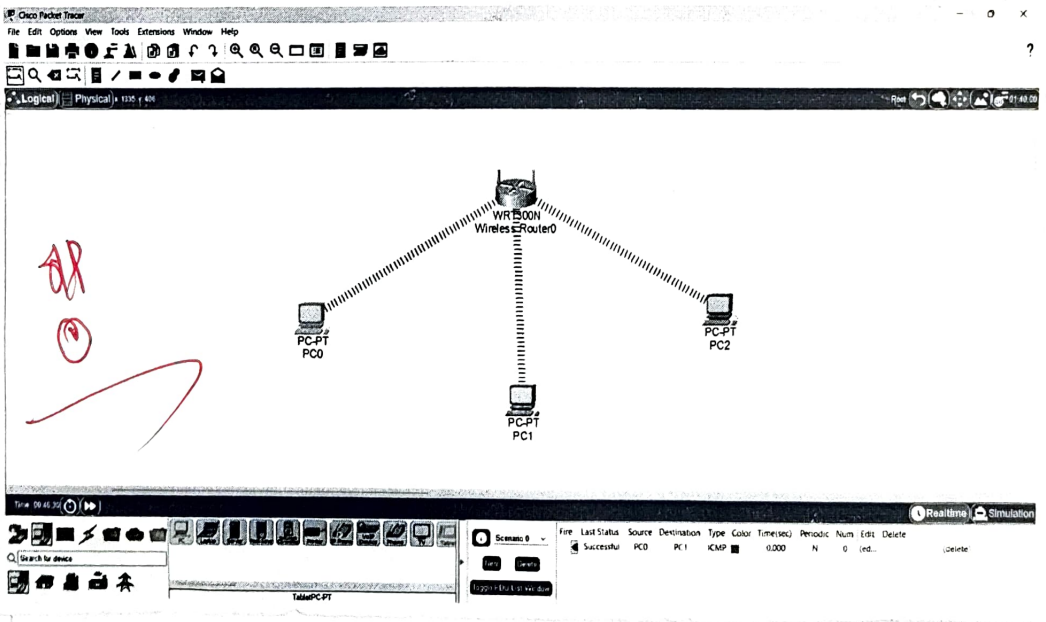
4) Configure the subnet mask as 255.255.255.0 & gateway as 192.168.0.1

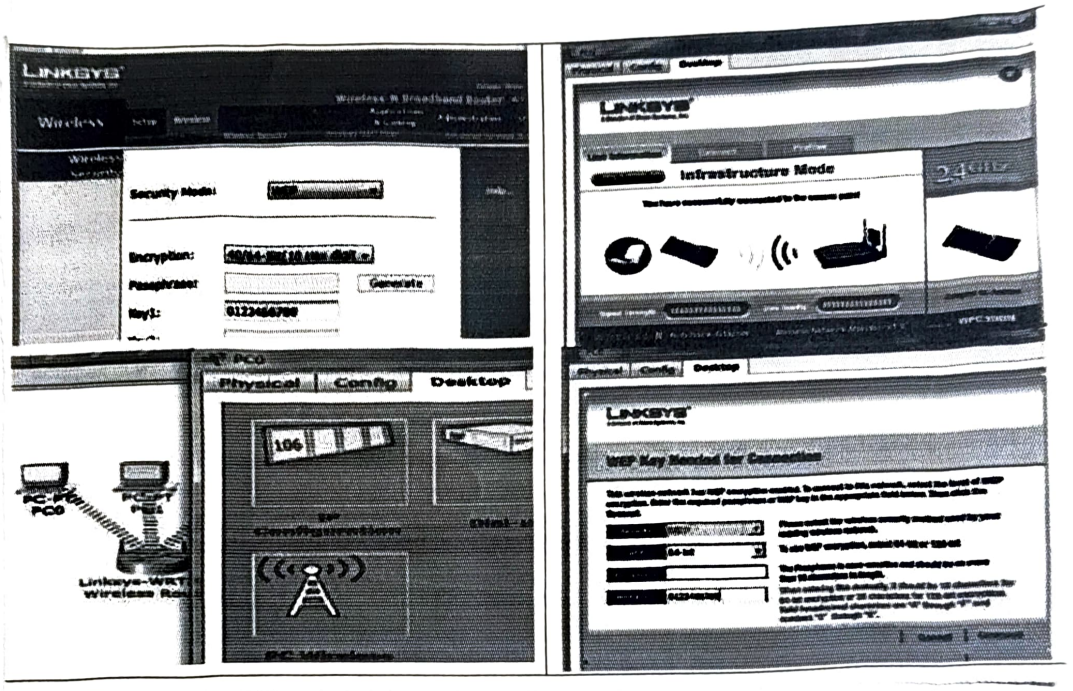
5) Connect all PC's to the wireless network using the WEP key 0123456789.

6) Use the ping command to test connectivity between the PCs.

7) Verify that all ping requests receives replies with 0% packet loss.


PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
C:\>
C:\>ping 192.168.0.3
Pinging 192.168.0.3 with 32 bytes of data:
Reply from 192.168.0.3: bytes=32 time=64ms TTL=128
Reply from 192.168.0.3: bytes=32 time=16ms TTL=128
Reply from 192.168.0.3: bytes=32 time=28ms TTL=128
Reply from 192.168.0.3: bytes=32 time=32ms TTL=128
Ping statistics for 192.168.0.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 16ms, Maximum = 64ms, Average = 32ms
C:\>
C:\>ping 192.168.0.4
Pinging 192.168.0.4 with 32 bytes of data:
Reply from 192.168.0.4: bytes=32 time=49ms TTL=128
Reply from 192.168.0.4: bytes=32 time=16ms TTL=128
Reply from 192.168.0.4: bytes=32 time=20ms TTL=128
Reply from 192.168.0.4: bytes=32 time=15ms TTL=128
Ping statistics for 192.168.0.4:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 15ms, Maximum = 49ms, Average = 25ms
C:\>
C:\>ping 192.168.0.1
Pinging 192.168.0.1 with 32 bytes of data:
Reply from 192.168.0.1: bytes=32 time=97ms TTL=255
Reply from 192.168.0.1: bytes=32 time=11ms TTL=255
Reply from 192.168.0.1: bytes=32 time=14ms TTL=255
Reply from 192.168.0.1: bytes=32 time=28ms TTL=255
Ping statistics for 192.168.0.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 11ms, Maximum = 97ms, Average = 22ms
C:\>
Top


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
WRT300N
Wireless-Router0
PC-PT
PC0
PC-PT
PC1
PC-PT
PC2
Realtime
Simulation
Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete
Successful PC0 PC1 ICMP 0.000 N 0 (ed... Delete



Result:-

Wireless LAN was configured successfully & connectivity was verified with 0% Packet loss